

TOKENHARVEST

Tea Marketplace

Powered by eWR Infrastructure

API Integration Reference Guide

Complete technical documentation for all integration roles:
Factory Portals, Producers, Brokers, Off-Takers,
Financiers, Cooperatives, and Exchange Administrators.

REST / JSON

Clerk JWT Auth

API Key (Service)

eWR Registry Sync

Version 1.0

Contact exchange@tokenharvest.io

Table of Contents

1	Base URL & Environments
2	Authentication Methods
3	User Tiers & Roles
4	Users & Profiles API
5	Electronic Warehouse Receipts (eWRs)
6	Tea Lots, Auctions & Bidding
7	Factory Portal Integration
8	Broker Mandates
9	Financing & Lending
10	Forward Contracts
11	Settlements & Disbursement
12	Cooperative Endpoints
13	Admin, Statistics & Registry
A	Appendix — Enumerations & Error Codes

Integration Role Quick Reference

Role / System	Auth Method	Primary Actions
Factory Portal	X-API-Key header	Push TEA eWRs to the marketplace
PRODUCER	Clerk JWT	View eWRs, self-list lots, request financing, forward contracts
ENABLER (Broker)	Clerk JWT	Receive mandates, catalogue lots, submit auctions, settle trades
OFF_TAKER	Clerk JWT	Browse market, place bids, buy fixed-price lots, sign forward contracts
FINANCIER	Clerk JWT	Approve/disburse loans, view loan book, audit platform earnings
COOPERATIVE	Clerk JWT	Aggregate member deliveries, request bulk eWRs, split eWRs
ADMIN	Clerk JWT	Full platform access — auctions, users, settlements, earnings
WRSC Registry	eWRS JWT / HMAC	Registry webhooks, lien sync, master reference data

1 Base URL & Environments

All API endpoints are served under the /api prefix relative to your deployment domain.

```
Production:    https://<your-domain>/api
Development:   https://<replit-dev-domain>/tea/api
```

Path prefix in development: The marketplace is mounted at the /tea/ path in the Replit preview environment. External systems targeting production do not need this prefix.

Global Request Headers

Header	Value	Required When
Authorization	Bearer <clerk-jwt>	All Clerk-authenticated endpoints
X-API-Key	<factory-api-key>	Factory service-to-service endpoints only
Content-Type	application/json	All POST / PATCH requests
Accept	application/json	Recommended on all requests

Error Response Envelope

All errors return a JSON object. Successful responses return the resource directly (not wrapped).

```
{
  "error": "Human-readable message describing the failure",
  "issues": [ /* Zod validation issues array – present on 400 validation errors */ ]
}
```

2.1 Clerk JWT (Standard User Auth)

The primary authentication mechanism for all human users. Obtain a session token via the Clerk SDK. Pass it as a Bearer token in the Authorization header on every request.

Client Platform	SDK / Method
Browser / SPA (React)	@clerk/react — useAuth().getToken()
Mobile (React Native)	@clerk/expo SDK
Server-to-server	Clerk Backend API — create a machine token
Postman / curl testing	Copy session token from browser DevTools !' Network tab

Request the Clerk publishable key from your exchange operator. Keys are environment-specific. Never embed secret keys in client code. Token lifetimes are managed by Clerk automatically.

2.2 API Key Auth (Factory / Service-to-Service)

Used exclusively by the factory portal. A shared secret is set as FACTORY_API_KEY on the API server and is passed in the X-API-Key header. No Clerk account is required.

```
# Example – push an eWR from factory system
curl -X POST https://<domain>/api/factory/ewrs \
  -H "X-API-Key: YOUR_FACTORY_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{ "factoryMark": "LIMURU_01", "ewrsReceiptId": "EWR-2026-00001", ... }'
```

Security: The /factory/ewrs endpoint is not mounted at all unless FACTORY_API_KEY is explicitly set on the server. Rotate the key immediately if compromised. All pushes share a single key per configured factory.

2.3 eWRS Registry JWT

Used by the Warehouse Receipt System Controller (WRSC) for registry webhooks and master data access. Requires WRSC_SECRET, EWR_JWT_SECRET, and EWR_OAUTH_CLIENTS to all be configured. This surface is completely disabled unless all three secrets are present. Contact the exchange operator to configure.

3

User Tiers & Roles

Every registered user is assigned exactly one tier at account creation. The tier is immutable after registration (except via Admin override). All endpoints enforce tier-based access server-side — the UI is a convenience layer only.

PRODUCER Tea estates, smallholder farmers, agricultural producers.	CAN DO Submit eWRs · Self-list lots (direct sales) · Request financing · Create forward contracts · Grant mandates to brokers RESTRICTIONS Cannot place bids, create auctions, or disburse settlements.
ENABLER (Broker) Licensed tea brokers and intermediaries.	CAN DO Receive mandates · Catalogue lots · Submit lots to auctions · Accept below-reserve bids · Manage dispatch documents · Disburse settlement legs RESTRICTIONS Cannot own eWRs or place bids as a buyer.
OFF_TAKER Tea buyers, blenders, packers, exporters.	CAN DO Browse spot market · Place auction bids · Purchase fixed-price lots · Sign forward contracts · Place spot orders RESTRICTIONS Cannot list lots, create mandates, or access admin.
FINANCIER Banks, microfinance institutions, trade finance providers.	CAN DO View eligible eWRs · Approve / disburse loans · View full loan book · Access earnings & audit logs RESTRICTIONS Cannot list lots, place bids, or create mandates.
COOPERATIVE Farmer cooperative societies and aggregators.	CAN DO Aggregate member deliveries into macro-lots · Request bulk eWRs · Split eWRs by member weight RESTRICTIONS Trades through member producers, not directly.
ADMIN Exchange operators and platform staff.	CAN DO Full access — create auction sessions, manage users, override tiers, oversee all settlements, view platform earnings RESTRICTIONS No restrictions — handle with care.

First-Time Registration Flow

When a user signs in for the first time, the frontend calls GET /users/me. If no DB record exists, the user completes their profile via PATCH /users/me (name, company, tier). Tier selection is permanent — warn users before they confirm. After registration, the tier cannot be changed without Admin intervention.

4

Users & Profiles API

GET

/users/me

Clerk JWT

Returns the authenticated user's full profile. On first login for super-admin accounts, auto-creates the DB record by matching the Clerk email against the configured admin email. All other users must call PATCH /users/me first.

ROLES

Any authenticated

RETURNS

id, clerkId, name, company, phone, nationalId, tier, reputationScore, kybStatus, createdAt

PATCH

/users/me

Clerk JWT

Updates the current user's profile. Tier can only be set once — if a tier is already recorded it is ignored in subsequent calls.

Field	Type	Constraint
name	string	optional
company	string	optional
phone	string	optional
nationalId	string	optional
tier	enum	OPTIONAL — immutable after first set. Values: PRODUCER ENABLER OFF_TAKER FINANCIER COOPERATIVE

GET

/users/brokers

Clerk JWT

Returns all users with the ENABLER tier. Used by producers to populate the broker picker when granting

mandates.

RETURNS

Array of { id,
name,
company }

GET

/users/:userId

Clerk JWT

Returns the public profile of any user by internal DB ID. Use for displaying counterparty information in trades.

RETURNS

{ id, name,
company, tier, re
putationScore }

eWR Lifecycle: INGESTED !' MARKET_LISTED !' AUCTION_ACTIVE !' SETTLED / EXTINGUISHED
Side states: ENCUMBERED (active lien/loan) · FORWARD_BOUND (pledged to a forward contract)
Terminal states: SETTLED and EXTINGUISHED cannot be listed, financed, or transferred.

POST /ewrs

Clerk JWT

Intakes a new eWR through the marketplace UI. Validates commodity-specific grading standards (EAS moisture limits for grain, cold-chain requirements for avocado). For grain, assigns to an existing pool group or creates a new one.

ROLES

PRODUCER

ENABLER

Field	Type	Requirement	Notes
commodityType	enum	REQUIRED	MAIZE RICE COFFEE TEA AVOCADO
ewrsReceiptId	string	REQUIRED	Exchange-issued receipt number — must be unique
warehouseCode	string	REQUIRED	WRSC-assigned facility code
weightMt	number	REQUIRED	Weight in metric tonnes
grade	string	REQUIRED	Commodity grade e.g. BOP (tea), Grade 1 (maize)
harvestSeason	string	REQUIRED	e.g. '2026A'
estimatedValueUsd	number	optional	
teaProcessingType	string	TEA only	CTC Orthodox
teaLeafGrade	string	TEA only	BOP BOPF PF BP D F
teaInvoiceSerial	string	TEA only	Factory invoice serial number

Status	Error Condition
422	Grading standards not met (moisture / cold-chain)

Status	Error Condition
409	Duplicate ewrsReceiptId
403	Caller tier cannot submit eWRs

GET	/ewrs/my-portfolio	Clerk JWT	Returns all eWRs owned by the caller, enriched with active lien/loan status, portfolio total value, and counts by state and commodity type. RETURNS { ewrs[], totalValueUsd, stateCounts{}, c ommodityCount s{} }
GET	/ewrs/broker-available	Clerk JWT	Returns TEA eWRs in INGESTED state owned by producers who have granted the calling broker an active mandate. Used to populate the eWR picker when a broker creates a new lot. ROLES
ENABLER			
POST	/ewrs/:ewrId/split	Clerk JWT	Splits one eWR into two by weight. The original is extinguished; two child eWRs are minted. eWR must have no active lien and be in INGESTED or MARKET_LISTED state.

Body Field	Type	Notes
------------	------	-------

Body Field	Type	Notes
splitWeightMt	number	REQUIRED — weight assigned to child A; remainder forms child B

RETURNS
{ original, childA, childB } — all three eWR objects

POST

/ewrs/:ewrId/transfer

Clerk JWT

Transfers eWR ownership to another registered user. No active lien may exist. For OTC title transfers outside the auction/order flow.

Body Field	Type	Notes
toUserId	integer	REQUIRED — target user's internal DB ID

POST

/ewrs/:ewrId/retire

Clerk JWT

Permanently extinguishes an eWR (state !' EXTINGUISHED). Used when goods are consumed, destroyed, or removed from the warehouse system. Irreversible — no active lien may exist.

Lot Status Flow: DRAFT !' CATALOGUED !' DISPATCHED !' LIVE !' SOLD

Also: DRAFT !' CATALOGUED !' SOLD (fixed-price, no auction needed)

Terminal: SOLD · UNSOLD · WITHDRAWN · RESERVE_NOT_MET

Creating a lot from an eWR immediately advances the eWR from INGESTED !' MARKET_LISTED.

6.1 Tea Lot Management

POST /tea/lots

Clerk JWT

Creates a tea lot catalogue entry (status DRAFT). Brokers (ENABLER) require an active mandate from the eWR owner. Producers (PRODUCER) self-list their own eWRs with zero commission — no mandate needed. The source eWR must be INGESTED. Only one active (non-WITHDRAWN) lot may exist per eWR.

ROLES

ENABLER

PRODUCER

Field	Type	Requirement	Notes
<code>ewrId</code>	integer	REQUIRED	Source eWR — must be INGESTED, no existing active lot
<code>grade</code>	string	REQUIRED	e.g. 'BOP'
<code>gradeMark</code>	string	REQUIRED	Estate / factory mark

Field	Type	Requirement	Notes
giOrigin	string	REQUIRED	e.g. 'Kenya'
grossWeightKg	number	REQUIRED	
netWeightKg	number	REQUIRED	
tareWeightKg	number	REQUIRED	
packageType	string	REQUIRED	e.g. 'Paper Sack'
listingType	enum	REQUIRED	AUCTION FIXED_PRICE
reservePriceUsd	number	If AUCTION	Price per kg — reserve floor
fixedPricePerKgUsd	number	If FIXED_PRICE	Price per kg — fixed sale price
certifications	string[]	optional	Default: []
tasterRemarks	string	optional	
commissionRate	decimal 0–1	optional	Default 0.01 (1%). Force-set to 0 for PRODUCER self-listings
bidSecurityPct	decimal 0–1	optional	Default 0.10 (10% bid security hold)

Status	Error Condition
403	Broker has no active mandate from eWR owner / Producer does not own the eWR
409	eWR not in INGESTED state or active lot already exists for this eWR

GET	/tea/lots	Clerk JWT
Filterable catalogue of all tea lots. Returns all columns plus warehouseCode (from the linked eWR) on each row.		

Query Param	Values	Notes
grade	string	Exact match
giOrigin	string	Exact match
listingType	AUCTION FIXED_PRICE	
status	DRAFT CATALOGUED DISPATCHED LIVE SOLD UNSOLD WITHDRAWN RESERVE_NOT_MET	
brokerId	integer	Filter by broker user ID
ownerId	integer	Filter by producer user ID
certification	string	JSONB array contains check

GET

/tea/lots/:id

Clerk JWT

Full lot detail including enriched eWR data (processing type, leaf grade, harvest season, warehouse code), owner/broker names, and best-effort warehouse operator profile lookup (matched on WRSC licence number).

RETURNS
lot + warehouse Code + wareh useProfile? + ownerName + brokerName + ewr (sub-object)

PATCH

/tea/lots/:id

Clerk JWT

Updates a DRAFT or CATALOGUED lot. Any PATCH on a DRAFT lot automatically advances it to CATALOGUED (published). Commission rate is locked at 0 for direct (producer self-listed) lots — the server enforces this regardless of what is sent. Only the mandate broker or eWR owner may call this.

ROLES

ENABLER	PRODUCER
Status	Error Condition
400	Lot status is not DRAFT or CATALOGUED
403	Caller is neither the mandate broker nor the eWR owner

POST

/tea/lots/:id/dispatch

Clerk JWT

Attaches a dispatch document. Attaching PRE_AUCTION_DISPATCH to a CATALOGUED lot auto-advances it to DISPATCHED. DELIVERY_ORDER can only be attached after the lot is SOLD.

Body Field	Values	Notes
docType	PRE_AUCTION_DISPATCH WEIGHMENT_REPORT DELIVERY_ORDER	REQUIRED
docData	object	OPTIONAL — arbitrary fields, e.g. { fileUrl: '...' }

6.2 Auction Sessions

POST	/tea/auctions	Clerk JWT	Creates a SCHEDULED auction session. Brokers then assign lots before the session goes live. ROLES
ADMIN	BODY auctionDate (string, YYYY-MM-DD) REQUIRED		
POST	/tea/auctions/:id/lots	Clerk JWT	Assigns CATALOGUED or DISPATCHED lots to a SCHEDULED session. The broker must be the mandate holder for each lot submitted. ROLES
ENABLER	ADMIN BODY lotIds: integer[] REQUIRED		
POST	/tea/auctions/:id/start	Clerk JWT	Starts the

auction.
 Session !' LIVE,
 first lot
 countdown
 begins.
 Requires at
 least one lot
 assigned.
 ROLES

ADMIN

BODY
 durationMins: integer (1–120, default 7) OPTIONAL

6.3 Bidding

POST /tea/lots/:id/bids

Clerk JWT

Places a bid on
 a LIVE lot.
 Enforces
 minimum bid
 increments (tick
 rules). Anti-
 snipe logic:
 bids within the
 configured
 window (default
 3 min) extend
 the timer. A bid
 security hold is
 recorded for the
 bidder (default
 10% of bid
 value × net
 weight).

ROLES

OFF_TAKER

BODY
 amountUsd: number REQUIRED — bid price per kg

Status	Error Condition
400	Bid is below minimum tick increment
400	Auction countdown has ended
403	eWR owner cannot bid on their own lot

POST /tea/lots/:id/accept-below-reserve

Clerk JWT

Accepts the
 highest bid on
 a RESERVE_N
 OT_MET lot.
 Triggers eWR
 ownership
 transfer and
 settlement
 record creation.

ROLES

ENABLER

ADMIN

GET

`/tea/lots/:id/settlement`



Returns the settlement breakdown for a specific lot. Restricted to parties in the trade and exchange admin.

ROLES

ADMIN

+ lot owner, mandate broker, and winning bidder

Service-to-service integration — no Clerk account required. Configure `FACTORY_API_KEY` on the API server and share the same value with the factory system. The route is not mounted until this variable is set. The producer must register on the marketplace and set their factory mark in their profile before any push will succeed.

How the factory push flow works

- 1 Producer registers on the marketplace and sets their factory mark (a unique string, e.g. "LIMURU_01") in their profile settings.
- 2 Factory system is configured with the shared `FACTORY_API_KEY` and the marketplace base URL.
- 3 After each batch is weighed and graded, the factory system calls `POST /factory/ewrs`.
- 4 The API looks up the producer by factory mark, mints the eWR in `INGESTED` state, and associates it with the producer's account.
- 5 The producer sees the new eWR in their dashboard and can immediately self-list it as a direct sale.

POST `/factory/ewrs`

X-API-Key

Ingests a TEA eWR from an external factory portal. The `factoryMark` is matched against `producer_profiles.factory_marks`. Idempotent on `ewrsReceiptId` — re-submitting an existing receipt ID returns the stored record without creating a duplicate.

Body Field	Type	Requirement	Notes
<code>factoryMark</code>	string	REQUIRED	Must match a registered producer profile
<code>ewrsReceiptId</code>	string	REQUIRED	Exchange receipt number — unique
<code>warehouseCode</code>	string	REQUIRED	WRSC-assigned facility code
<code>weightMt</code>	number	REQUIRED	Weight in metric tonnes
<code>grade</code>	string	REQUIRED	e.g. 'BOP'
<code>harvestSeason</code>	string	REQUIRED	e.g. '2026A'
<code>teaProcessingType</code>	string	optional	CTC Orthodox

Body Field	Type	Requirement	Notes
teaLeafGrade	string	optional	BOP BOPF PF BP D F
teaInvoiceSerial	string	optional	Factory invoice serial
estimatedValueUsd	number	optional	Estimated market value in USD

Status	Condition
401	Missing or invalid X-API-Key header
404	No producer profile found with this factoryMark
409	Duplicate ewrsReceiptId — existing record returned, no duplicate created

Request & Response Example

```
POST /api/factory/ewrs
X-API-Key: sk_factory_live_XXXXXXXXXXXXXXXXXX
```

```
{
  "factoryMark": "LIMURU_01",
  "ewrsReceiptId": "EWR-TEA-2026-00847",
  "warehouseCode": "WRSC-KE-NBI-014",
  "weightMt": 2.450,
  "grade": "BOP",
  "harvestSeason": "2026A",
  "teaProcessingType": "CTC",
  "teaLeafGrade": "BOPF",
  "teaInvoiceSerial": "INV-2026-0221",
  "estimatedValueUsd": 3200
}
```

201 Created response:

```
{
  "id": 147,
  "ewrsReceiptId": "EWR-TEA-2026-00847",
  "ownerId": 23,
  "commodityType": "TEA",
  "state": "INGESTED",
  "warehouseCode": "WRSC-KE-NBI-014",
  "weightMt": 2.450,
  "grade": "BOP",
  "harvestSeason": "2026A",
  "createdAt": "2026-07-26T08:15:00.000Z"
}
```

A mandate is formal permission granted by a commodity owner (PRODUCER or COOPERATIVE) to a broker (ENABLER) to act on their behalf for a specific commodity type. Brokers cannot catalogue lots without an active mandate from the eWR owner.

Only one active mandate per producer–broker–commodity combination may exist at a time. Attempting to create a duplicate returns HTTP 409.

POST **/broker-mandates**

Clerk JWT

Grants a mandate. The caller is the mandate owner. Commission rate override is stored as a decimal ratio (e.g. 0.05 = 5%) and applied automatically when the broker creates lots for this producer.

ROLES

PRODUCER

COOPERATIVE

Field	Type	Requirement	Notes
<code>brokerId</code>	integer	REQUIRED	ENABLER user ID — use GET /users/brokers to list
<code>commodityType</code>	enum	REQUIRED	MAIZE RICE COFFEE TEA AVOCADO
<code>commissionRateOverride</code>	decimal 0–1	optional	e.g. 0.03 for 3%
<code>permissions</code>	string[]	optional	Defaults to full set: list, accept_bids, negotiate, set_reserve
<code>validFrom</code>	ISO8601	optional	Mandate start date
<code>validTo</code>	ISO8601	optional	Mandate end date — open-ended if omitted

Status	Error Condition
403	Caller is not PRODUCER or COOPERATIVE
404	Broker user not found
409	Active mandate already exists for this producer–broker–commodity combination

GET **/broker-mandates/my**

Clerk JWT

Returns all mandates granted to the calling broker. Includes ownerName and active status.

ROLES

ENABLER

GET /broker-mandates/given

Clerk JWT

Returns all mandates granted by the calling producer or cooperative. Includes brokerName for display.

ROLES

PRODUCER

COOPERATIVE

DELETE /broker-mandates/:id

Clerk JWT

Revokes an active mandate (sets revoked = true). The broker can no longer create lots for this producer for this commodity. Existing lots created under this mandate are unaffected.

ROLES
Mandate owner (PRODUCER / COOPERATIVE)

Status	Error Condition
403	Caller is not the mandate owner
400	Mandate is already revoked

Producers pledge unencumbered eWRs as collateral to request warehouse-receipt financing. Financiers review, approve, and disburse loans. The eWR is locked (ENCUMBERED) for the loan duration and released upon settlement.

Eligibility: eWR must be in INGESTED or MARKET_LISTED state with no active lien (isLienActive = false).
Max loan-to-value: 60% of estimated value · Default interest rate: 12% per annum

GET /financing/eligible-ewrs

Clerk JWT

Returns eWRs eligible for financing. Producers see only their own. Financiers see all eligible eWRs platform-wide. Each item includes IMaxUsd (max loan = 60% of estimated value) and the default interest rate.

ROLES

PRODUCER

FINANCIER

ENABLER

POST /financing

Clerk JWT

Creates a PENDING financing request against an eligible eWR. Only one active request per eWR may exist at a time.

ROLES

PRODUCER

Body Field	Type	Notes
ewrId	integer	REQUIRED — must be eligible
notes	string	optional

Status	Error Condition
409	Active financing request already exists for this eWR
400	eWR state is ineligible (already encumbered or terminal)

GET /financing/:requestId

Clerk JWT

Returns request details enriched with

eWR metadata and linked loan data.

ROLES

PRODUCER FINANCIER ENABLER

PATCH /financing/:requestId/approve

Clerk JWT

Approves a PENDING request. Sets eWR !' ENCUMBERED, activates lien (isLienActive = true, lienHolderId = financier), creates a loan record, and simulates a WRSC lien lock.

ROLES

FINANCIER

PATCH /financing/:requestId/disburse

Clerk JWT

Confirms loan funds have been sent to the producer's bank account. Status !' DISBURSED. Simulates bank transfer and registry capital ingress recording.

ROLES

FINANCIER

PATCH /financing/:requestId/reject

Clerk JWT

Rejects a PENDING financing request. Status !' REJECTED. eWR remains unencumbered.

ROLES

FINANCIER

GET /financing/loan-book

Clerk JWT

Returns all active loans with live accrual calculations: accruedInterestUsd

and totalRepay
ableUsd based
on days
elapsed since
disbursement.
ROLES

FINANCIER

ENABLER

10 Forward Contracts

Forward contracts allow producers to lock in a future delivery price with a buyer before harvest or processing is complete. A 15% performance bond is required from both parties. The eWR is pledged (FORWARD_BOUND) for the contract duration.

POST /forwards

Clerk JWT

Creates a PENDING_SIGNATURE contract. The producer pledges an eWR and sets delivery price and maturity date. Status remains PENDING_SIGNATURE until the buyer co-signs.

ROLES

PRODUCER

Body Field	Type	Requirement	Notes
ewrId	integer	REQUIRED	Must be INGESTED or MARKET_LISTED
maturityDate	ISO8601	REQUIRED	Must be in the future
deliveryPriceUsd	number	REQUIRED	Total contract delivery value

RETURNS

Contract object including performanceBondUsd (15% of deliveryPriceUsd)

GET /forwards

Clerk JWT

Lists forward contracts. Filterable by status, sellerId, buyerId.

Query Param	Type	Notes
status	string	PENDING_SIGNATURE ACTIVE MATURED DEFAULTTED
sellerId	integer	Filter by producer
buyerId	integer	Filter by off-taker

POST /forwards/:contractId/co-sign

Clerk JWT

Buyer co-signs the contract. Status !' ACTIVE. eWR !' ENCUMBERED (lien to buyer). Performance bonds for both parties activated. eWR

cannot be
financed,
traded, or
transferred
while active.
ROLES

OFF_TAKER

POST /forwards/:contractId/complete

Clerk JWT

Triggers
maturity
resolution if the
maturity date
has passed.
Status !'
MATURED.
Performance
bonds released.
eWR remains E
NCUMBERED
pending full
settlement.
ROLES

ENABLER

PRODUCER

OFF_TAKER

POST /forwards/:contractId/resolve-default

Clerk JWT

Declares a
default. The
defaulting party
forfeits their
performance
bond and loses
10 reputation
points. Status !'
DEFAULTED.
eWR released
back to
INGESTED.
ROLES

ENABLER

+ SELLER (if declaring buyer default) or BUYER (if declaring seller default)

Body Field	Type	Notes
defaultSide	enum	REQUIRED — BUYER SELLER

The settlement engine handles the financial close-out for all trade types: auction lots, fixed-price orders, and forward contracts. It calculates platform fees, manages lien repayment, and orchestrates sequential payment leg disbursement.

Fee structure:

Platform fee: 2% of gross trade value (all trade types)

Escrow fee: 0.5% on spot orders

Financing facilitation: 0.5% of interest charged

Disbursement order: Bank leg (if lien active) !' Platform leg !' Producer leg

Completing the final leg transfers eWR title to the buyer and issues a signed Digital Release Token.

POST /settlements

Clerk JWT

Initiates settlement for a completed trade. Calculates all financial legs: platform fee (2%), bank repayment (from linked loan if lien active), and producer net payout. Sum of all legs equals gross trade value.

ROLES

ENABLER

FINANCIER

ADMIN

+ transaction parties (seller / buyer)

Body Field	Type	Requirement	Notes
entityType	enum	REQUIRED	ORDER AUCTION FORWARD
entityId	integer	REQUIRED	ID of the order, lot, or forward contract
loanId	integer	optional	Linked loan for mandatory lien repayment
notes	string	optional	

RETURNS

Settlement object with: bankLegUsd, platformFeeUsd, producerNetUsd, grossValueUsd, status

Status	Error Condition
409	Settlement already exists for this entity
400	Loan/eWR mismatch

GET /settlements/:settlementId

Clerk JWT

Returns settlement status,

calculated legs, linked loan details, and release token if settlement is complete. Restricted to involved parties and exchange admin.

ROLES

ADMIN **FINANCIER**

+ lot owner, mandate broker, and buyer

POST **/settlements/:settlementId/disburse** **Clerk JWT**

Confirms disbursement of a specific payment leg. Must follow the sequence: bank !' platform !' producer. Attempting to disburse out of order returns an error. On final leg completion, eWR ownership transfers to the buyer and a signed Digital Release Token is included in the response.

ROLES

ENABLER **FINANCIER** **ADMIN**

Body Field	Values	Notes
leg	bank platform producer	REQUIRED — must be disbursed in order

RETURNS
Updated settlement. On final leg: releaseToken (HMAC-signed) included in response body.

12 Cooperative Endpoints

Cooperatives aggregate member deliveries into macro-lots before minting a single bulk eWR. This allows smallholder farmers to pool their output while the cooperative manages the logistics.

GET /cooperatives/me

Clerk JWT

Returns the caller's cooperative profile and member list.

POST /cooperatives/me/macro-lots

Clerk JWT

Creates a macro-lot. Groups member deliveries by grade and harvest season.

POST /cooperatives/me/macro-lots/:id/finalise

Clerk JWT

Closes a macro-lot to new intake entries. Required before requesting an eWR.

POST /cooperatives/me/macro-lots/:id/request-ewr

Clerk JWT

Mints a fungible eWR for a finalised macro-lot. Body requires: warehouseCode (string).

**eWR Split
(used by cooperatives to divide lots by member)**

POST /ewrs/:ewrId/split

Clerk JWT

Splits a cooperative macro-lot eWR into two by weight — one for each member allocation. The

original eWR is extinguished and two child eWRs are created. No active lien may exist.

Body Field	Type	Notes
<code>splitWeightMt</code>	number	REQUIRED — weight of child A; remainder forms child B

RETURNS
{ original, childA, childB } — three eWR objects

13 Admin, Statistics & Registry

13.1 Administration

GET /admin/users

Clerk JWT

Lists and searches all registered users. Supports name / email / company filtering. Used for KYB management and tier overrides.
ROLES

ADMIN

ENABLER

FINANCIER

Query Param	Notes
search	Name / email / company substring search
tier	Filter by tier value
kybStatus	PENDING APPROVED REJECTED

PATCH /admin/users/:id

Clerk JWT

Updates a user's tier or KYB status. The only pathway to change a user's tier after registration. KYB status controls whether a user can transact at higher values.
ROLES

ADMIN

Body Field	Values	Notes
tier	enum	OPTIONAL — any valid tier
kybStatus	PENDING APPROVED REJECTED	optional

GET /admin/earnings

Clerk JWT

Platform earnings summary: total platform fees (2% of trade value), escrow

fees (0.5% on orders), financing facilitation fees (0.5% of interest charged).
ROLES

ADMIN ENABLER FINANCIER

RETURNS
{ platformFeesUsd, escrowFeesUsd, financingFacilitationUsd, totalUsd }

GET /audit

Clerk JWT

Returns audit log entries. All significant state changes produce audit entries. Filterable by entity type and entity ID.
ROLES

ADMIN ENABLER FINANCIER

13.2 Market Statistics

Endpoint	Returns
GET /stats/market-summary	Total value locked, trade volume, active lot and user counts
GET /stats/commodity-breakdown	eWR and volume stats grouped by commodity type
GET /stats/recent-activity	Feed of recent orders and settlements
GET /stats/market-risk	Caller's active liens and pending settlement totals
GET /stats/warehouse-distribution	Inventory counts and values grouped by warehouse code
GET /stats/price-trends	Clearing prices from historical auction lots

13.3 WRSC Registry Integration

Restricted surface — only mounted when WRSC_SECRET, EWR_JWT_SECRET, and EWR_OAUTH_CLIENTS are all configured. The WRSC team coordinates configuration directly with the exchange operator.

Endpoint	Auth	Purpose
GET /ewr/master/counties	eWRS JWT	Reference list of counties for intake forms
GET /ewr/master/commodity-types	eWRS JWT	Supported commodity types and grading standards
GET /ewr/master/harvest-seasons	eWRS JWT	Active and historical harvest seasons
POST /webhooks/registry-sync	HMAC signature	Inbound lien status updates and title transfer confirmations from the WRSC

13.4 File Storage

POST /storage/uploads/request-url

Clerk JWT

Returns a

presigned
upload URL for
object storage.
Use for
dispatch
documents,
warehouse
reports, or any
file attachment.

Body Field	Type	Notes
name	string	REQUIRED — filename
size	integer	REQUIRED — file size in bytes
contentType	string	REQUIRED — MIME type e.g. application/pdf

RETURNS
{ uploadUrl, objectPath } — PUT the file binary directly to uploadUrl,
then store objectPath in docData

eWR States

Value	Meaning	Terminal?
INGESTED	Freshly minted; available for listing or financing	No
MARKET_LISTED	An active lot exists against this eWR	No
AUCTION_ACTIVE	Currently live in an auction session	No
ENCUMBERED	Active lien (loan or forward contract)	No
FORWARD_BOUND	Pledged to a forward contract pre-co-sign	No
SETTLED	Title transferred to buyer	YES
EXTINGUISHED	Physically retired / destroyed	YES

Lot
Statuses

Value	Meaning
DRAFT	Created; not yet published to the catalogue
CATALOGUED	Published — visible in the marketplace catalogue
DISPATCHED	Pre-auction dispatch document attached
LIVE	Active in an auction session — bidding open
SOLD	Winning bid accepted or fixed-price purchased
UNSOLD	Auction closed with no bids placed
RESERVE_NOT_MET	Bids were placed but none reached the reserve
WITHDRAWN	Manually removed from the market

Commodity Types

Value	Grading Standard / Notes
TEA	Supported via factory push (POST /factory/ewrs) and direct UI intake
MAIZE	EAS 40 — moisture content < 13.5%
RICE	EAS 5 — moisture content < 14.5%
COFFEE	Moisture content < 12.5%
AVOCADO	Cold-chain requirements enforced; time-decay tracking enabled

HTTP Status Codes

Code	Meaning
200	OK — resource returned
201	Created — new resource created successfully
400	Bad Request — validation error in body (see issues[] array)
401	Unauthenticated — missing or invalid Bearer / API key
403	Forbidden — authenticated but insufficient tier or role
404	Not Found — resource does not exist
409	Conflict — duplicate resource or invalid state transition

Code	Meaning
422	Unprocessable Entity — grading / cold-chain standard not met
500	Internal Server Error — contact exchange operator

Listing Types

Value	Required Pricing Field	Settlement Path
AUCTION	reservePriceUsd (per kg)	Winning bid !' accept !' settle !' disburse
FIXED_PRICE	fixedPricePerKgUsd (per kg)	POST /orders !' settlement !' disburse